

Department of CSE

TUTORIAL PLAN

Name of the Course-Coordinator:

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Branch	Semester	Subject	Code	Session
CSE	3 rd	Computer Architecture	CCS103	July-Dec.2025

Tutorial Sheet No. 2

Course-Content Covered:

Addition, Subtraction, Multiplication and division algorithms and hardware, Register transfer language and operations, Arithmetic micro-operations, Logic micro-operations, Shift micro-operations, Arithmetic logic shift unit.

Question 1) Explain the following:

- a) A 4 bit ALU is designed to perform arithmetic, logical and shift micro operations. Given the following inputs: A = 1011, B = 1100. Perform the operations to provide resulting 4-bit output of each.

Question 2) Explain the following:

- a) Using Booths multiplication Algorithm multiply the following two signed 4 bit binary numbers in 2's complement representation.

Multiplicand (M) = 1011 (i.e, -5)

Multiplier (Q) = 1101 (i.e, -3)

- b) Perform binary addition of two numbers using standard binary addition algorithm and indicate if there is a carry out and overflow and provide the final result in binary and decimal.